

Article

A Wearable System for Real-Time Fall Detection on Resource-Constrained Devices

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Abstract

In this study, we propose a wearable fall detection system that combines wearable sensors, TinyML model, and IoT-based communication for real-time monitoring and detection of falls. The system is designed for resource-constrained IoT devices where memory, power, and processing capacity are limited. The system works by collecting body motion data using accelerometer sensors placed on the human body. The data is then processed using a feedforward neural network trained on preprocessed signals. The trained model is quantized so that it can run on low-power embedded hardware with small memory size. The model performs inference directly on the device. This reduces latency and avoids sending raw sensor data to the cloud. When a fall is detected, the result is sent through Bluetooth to a gateway. The gateway forwards the data to a cloud server using the MQTT protocol. The cloud stores the data and supports monitoring and analysis. The experimental results show that the quantized TinyML model achieves 98.40% accuracy with more than 80% F1-score and more than 99% recall. The deployed model uses only ~5 KB of RAM and ~40 KB of flash memory. The inference time is 7 ms per class. These results show that wearable sensing with quantized TinyML models and IoT communication can provide fast and reliable fall detection for real-world safety monitoring systems.

Keywords: falldetection; human activity recognition; internet of things (IoT); smart healthcare; wearable sensors; industry innovation and infrastructure; good health and well-being



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1. Introduction

Human Activity Recognition is a growing research field that focuses on developing machine learning models to detect and classify different human activities. The activity recognition system is used in many domains such as healthcare, sports training, and child monitoring. Traditional activity detection systems use fixed sensors in the environment or sensors attached to different parts of the human body. These setups can be uncomfortable and inconvenient for the person who uses them. The progress in TinyML and embedded machine learning have now enabled machine learning models for activity recognition to run directly on resource-constrained microcontrollers, which as a result reduces latency, bandwidth for sending and receiving data, and dependence on cloud inference [1–3].